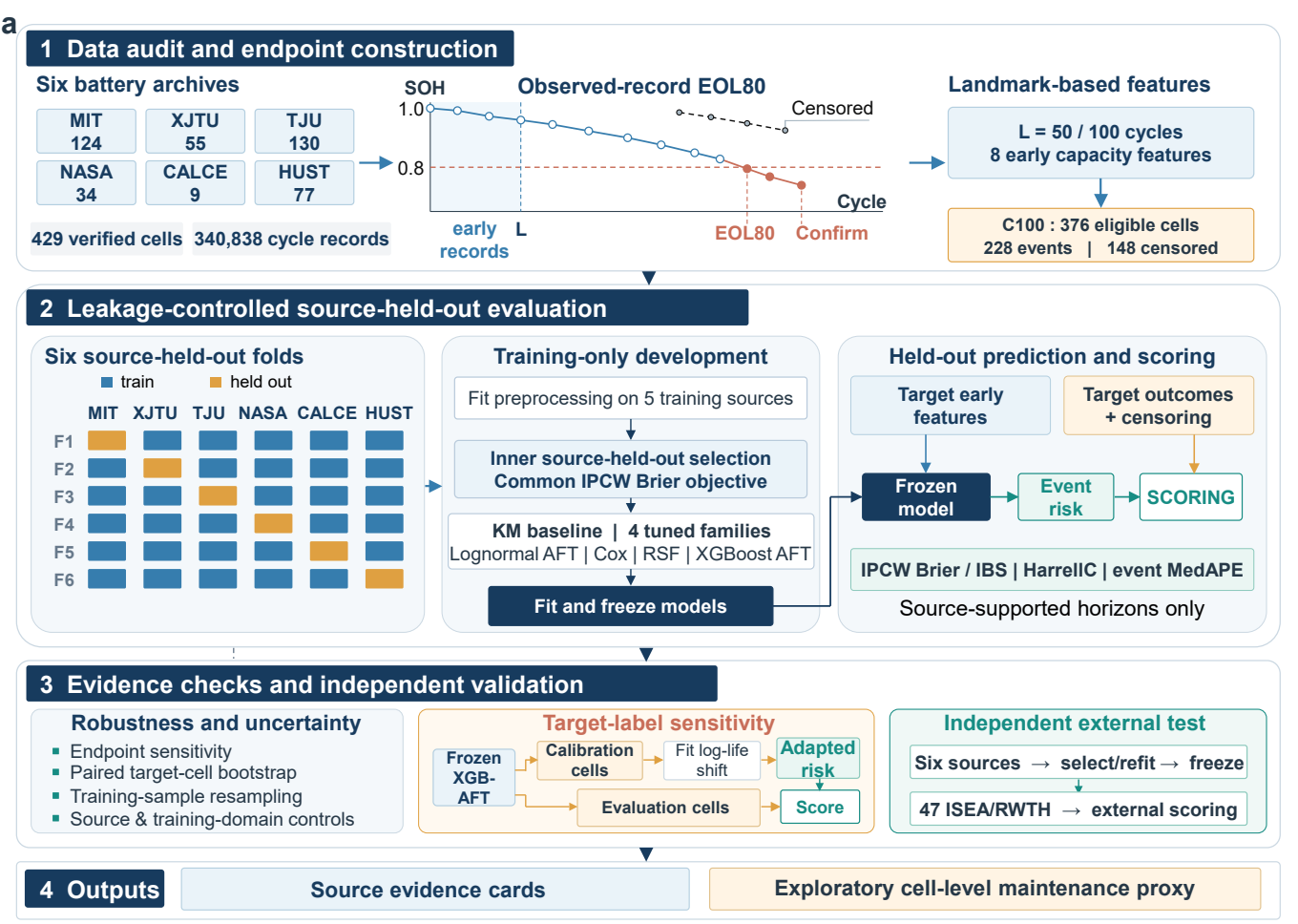
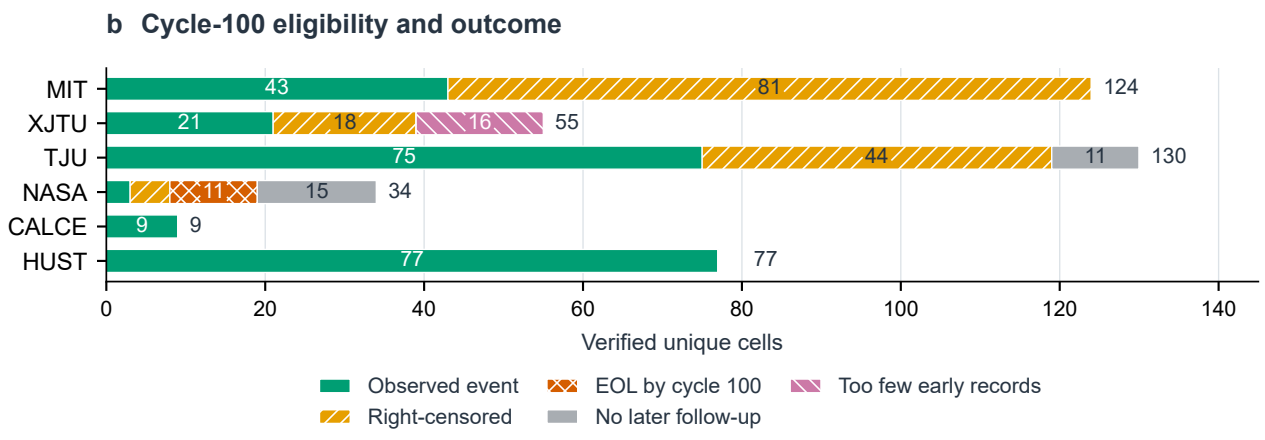

a
1 Data audit and endpoint construction
Six battery archives
MIT 124
XJTU 55
TJU 130
NASA 34
CALCE 9
HUST 77
429 verified cells
340,838 cycle records
SOH
1.0
0.8
Observed-record EOL80
Censored
Cycle
early records
L
EOL80
Confirm
Landmark-based features
L = 50 / 100 cycles
8 early capacity features
C100 : 376 eligible cells
228 events | 148 censored
2 Leakage-controlled source-held-out evaluation
Six source-held-out folds
train
held out
MIT XJTU TJU NASA CALCE HUST
F1
F2
F3
F4
F5
F6
Training-only development
Fit preprocessing on 5 training sources
Inner source-held-out selection
Common IPCW Brier objective
KM baseline | 4 tuned families
Lognormal AFT | Cox | RSF | XGBoost AFT
Fit and freeze models
Held-out prediction and scoring
Target early features
Target outcomes + censoring
Frozen model
Event risk
SCORING
IPCW Brier / IBS | HarrellC | event MedAPE
Source-supported horizons only
3 Evidence checks and independent validation
Robustness and uncertainty
Endpoint sensitivity
Paired target-cell bootstrap
Training-sample resampling
Source & training-domain controls
Target-label sensitivity
Frozen XGB-AFT
Calibration cells
Fit log-life shift
Adapted risk
Evaluation cells
Score
Independent external test
Six sources → select/refit → freeze
47 ISEA/RWTH → external scoring
4 Outputs
Source evidence cards
Exploratory cell-level maintenance proxy


b Cycle-100 eligibility and outcome
MIT
XJTU
TJU
NASA
CALCE
HUST
43
81
124
21
18
16
55
75
44
11
130
11
15
34
9
9
77
77
0
20
40
60
80
100
120
140
Verified unique cells
Observed event
Right-censored
EOL by cycle 100
No later follow-up
Too few early records